

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 08)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

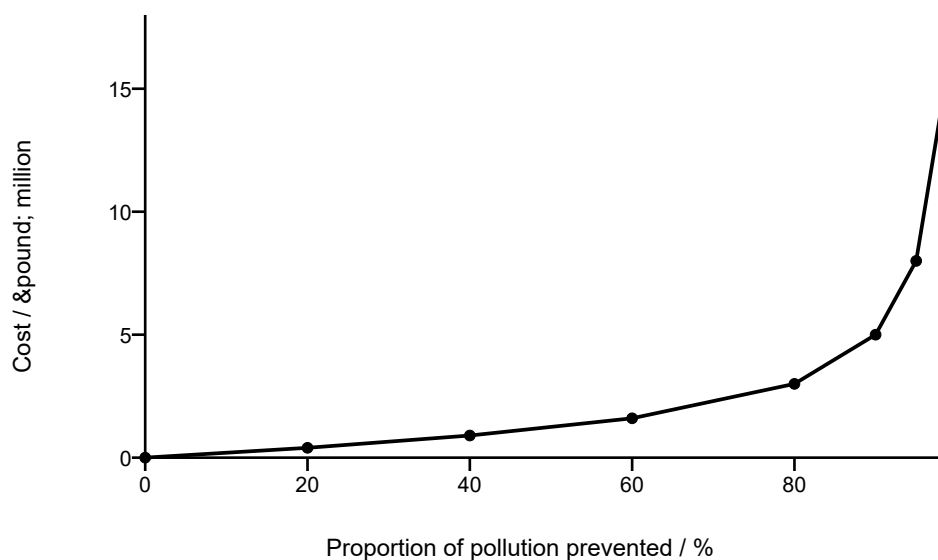
Question	Mark
1	
2	
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7	
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9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

Figure 1 shows how the cost of controlling emissions from a factory changes with the proportion of the pollutant that is prevented from being released.

Figure 1



0 1

1

Use Figure 1 to calculate the extra cost of increasing control from 90% to 99% of the pollutant.

[1 mark]

Answer

£ million

0 1

2

Explain how the terms **ALARA** and **BATNEEC** are used when deciding how much pollution control is required.

[3 marks]

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0 1 . 3

Explain why the control methods that were originally adequate for a factory may no longer be sufficient.

[2 marks]

0 1 . 4

Evaluate the principles and general strategies that can be used to decide how pollution should be controlled.

[9 marks]

Do not write outside the box

Extra space

15

0 2 . 1

Which of the following statements about stratospheric ozone is correct? Tick one box.

[1 mark]

- A Ozone is destroyed only by CFCs ☐
- B Ozone absorbs mainly infrared radiation ☐
- C Ozone is formed when monatomic oxygen reacts with diatomic oxygen ☐
- D Ozone concentrations are highest at ground level ☐

0 2 . 2

Explain the difference between the roles of ozone in the stratosphere and in the troposphere.

[2 marks]

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0 2 . 3

Explain why HCFCs cause less ozone depletion than CFCs.

[2 marks]

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5

0 3 . 1

Complete Table 1 to explain how each method reduces soil erosion.

[5 marks]

Table 1

Method	How it reduces erosion	
Terracing		
Tied ridging		
Windbreaks / hedgerows		
Mulching		
Strip cropping		

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0 3 . 2 Explain why long-term crops such as tea, coffee or orchards reduce erosion compared with annual arable crops. **[2 marks]**

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0 3 . 3 Describe **three** ways livestock can be managed to reduce soil erosion. **[3 marks]**

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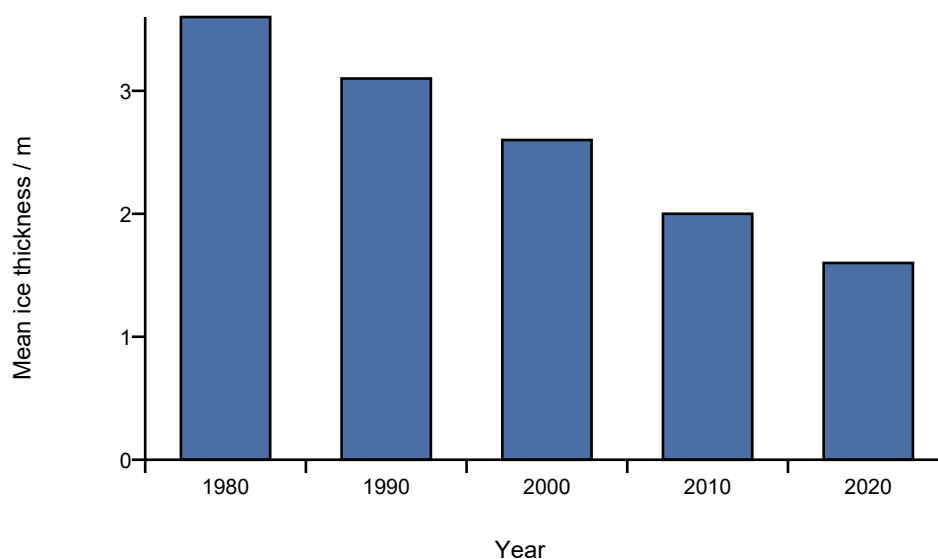
10

0 4

Table 1 shows the area of Arctic sea ice in September (the annual minimum) in selected years. Figure 1 shows the mean thickness of Arctic sea ice in the same years.

Table 1

Year	1980	1990	2000	2010	2020
September sea ice area / million km ²	7.7	6.2	6.3	4.9	3.9

Figure 1**0 4****1**

Calculate the percentage decrease in September sea ice area between 1980 and 2020.

[2 marks]**Answer**

%

0 4**2**

Explain why the loss of sea ice does **not** directly raise sea level, but the loss of land ice does.

[2 marks]

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0 4 . 3

Explain how the changes shown in Table 1 and Figure 1 could lead to further warming.

[3 marks]

0 4 . 4

Explain **two** ways in which rising temperatures cause sea level to rise.

[2 marks]

0 4 . 5

Explain why the break-up of an ice shelf can increase the rate at which land ice flows into the sea.

[1 mark]

10

0 5 . 1

Which of the following is the main advantage of composting organic waste rather than landfilling it? Tick **one** box.

[1 mark]

- A Aerobic decomposition produces little methane and returns nutrients ☐ to the soil
- B It produces more electricity ☐
- C It removes heavy metals ☐
- D It destroys asbestos fibres ☐

0 5 . 2

Explain why the ash from waste incinerators must be disposed of carefully.

[2 marks]

0 5 . 3

Give **two** features of a well-managed landfill site that reduce pollution.**[2 marks]**

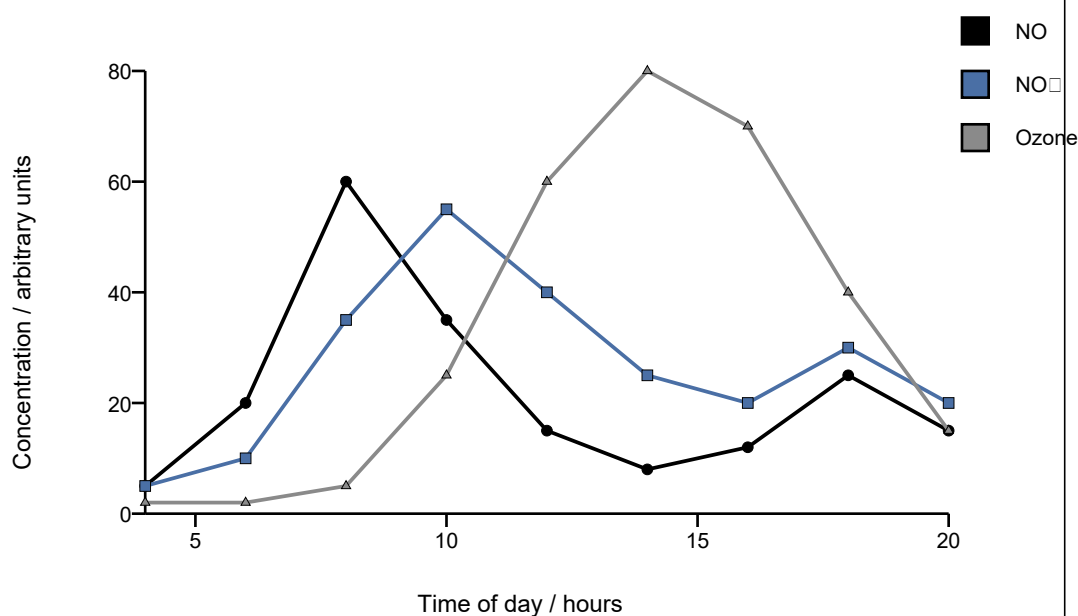
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0 6

Figure 1 shows the concentrations of three pollutants in the air of a large city in a valley during one sunny day.

Figure 1

0 6 . 1

Use Figure 1 to describe how the ozone concentration changes during the day and explain the pattern.

[4 marks]

06 . 2

Explain why the early-morning peak in NO is followed by a peak in NO₂.

[2 marks]

06 . 3

Name **one** other toxic secondary pollutant produced in photochemical smogs and give one of its effects on humans.

[2 marks]

06 . 4

Explain why cities such as Los Angeles and Mexico City are particularly prone to photochemical smogs.

[2 marks]

10

07 . 1

Explain why mineral extraction causes more land-use conflict than most other human activities.

[2 marks]

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07 . 2

Describe how each of the following environmental impacts of an operating mine can be reduced.

[5 marks]

Dust

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Noise

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Turbid drainage water

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Acidic leachate from spoil

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Loss of amenity

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07 . 3

Explain how spoil heaps can be made stable and describe what happened at Aberfan in 1966.

[3 marks]

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Evaluate the risks and benefits of the uses of ionising radiation, and explain the principles used to control exposure of workers and the public.

[9 marks]

Extra space

08 . 2 Give **one** reason why the risks and benefits of nuclear electricity may not fall on the same people.

[1 mark]

10

09 . 1

Which of the following water sources is usually the most convenient for large communities? Tick one box.

[1 mark]

A Desalinated seawater

☐

B River water

☐

C Rainwater catchment

☐

D Newly discovered aquifers

☐

09 . 2

Explain why the quality requirements for water differ between uses, using **two** named uses.

[2 marks]

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09 . 3

Explain why inter-basin transfers may be used and give **one** disadvantage.

[2 marks]

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10

A student investigated whether the number of earthworms per m^2 differs between an organic field and a conventionally farmed field. Table 1 shows the results.

Table 1

Field	Number of samples	Mean earthworms per m^2	Standard deviation
Organic	10	48	14
Conventional	10	31	12

10 . 1

Name the statistical test the student should use and explain why it is appropriate.

[2 marks]

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1 0 . 2

The test gave $p = 0.01$. State the conclusion the student should draw.

[2 marks]

1 0 . 3

Explain why the student cannot conclude that organic farming **caused** the higher earthworm numbers.

[2 marks]

1 0 . 4

Evaluate the decisions that must be made when planning a fieldwork investigation so that valid conclusions can be drawn.

[9 marks]

Do not write outside the box

Extra space

Do not write outside the box

15

Do not write outside the box

Write an essay on one of the following topics.

1	1	.	1
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Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.

[25 marks]

OR

1 1 . 2

Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS